



Combined effect of rice husk ash and copper slag on the mechanical performance of cement mortar

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ABSTRACT

The growing demand for sustainable and high-performance construction materials has driven extensive research into the utilisation of industrial and agricultural by-products as substitutes in mortar and concrete. At the same time, increased industrial and agricultural waste has created disposal management issues at the local and global levels. To address this issue, the infrastructure sector needs to play a big role in minimising the ill effects of agro-industrial wastes on the environment. Rice husk ash (RHA), an agricultural and a highly reactive pozzolanic material, and copper slag (CS), a by-product of the copper smelting process, show significant potential for incorporation into cementitious systems and therefore this study examines the effects of partial replacement of cement with RHA (10 % and 20 %) and manufactured fine aggregates with CS (10 %, 30 % and 50 %) on the mechanical properties of cement mortar. Experimental investigations included flowability, compressive strength, flexural strength, and ultrasonic pulse velocity (UPV) tests. The results indicate that the combination of 10 % RHA and 50 % CS produces the most favourable compressive and flexural strength performance. The UPV tests show RHA and CS matrix improve the cementitious matrix at all various replacement levels. The UPV correlation with flexural strength at seven and twenty-eight days showed moderate correlation of R^2 values of 0.48 and 0.42, respectively. Furthermore, to understand the effects of replacement materials and to obtain a simple expression for compressive and flexural strength, multiple regression analysis was employed to develop predictive models correlating flexural and compressive strengths with influencing factors such as curing age, RHA content, CS content, and UPV. From the results obtained, it was identified that the overall correlation was higher for both models, with an R^2 value of 0.73 and 0.76. An Image-based cross-sectional analysis using ImageJ confirmed the random distribution of CS particles within the mortar matrix, with grey value–pixel plots validating its heterogeneous dispersion for all the mixes, especially the matrix with the highest replacement level.

1. Introduction

The ever-increasing demand for fine and coarse aggregates in the construction industry has raised significant ecological concerns, as the continuous extraction of these natural resources contributes to environmental degradation and resource depletion. Concurrently, the rapid expansion of ore-based industries—such as those involved in the production of iron, aluminium, and copper—has generated substantial quantities of industrial by-products that are frequently disposed of in

landfills, posing risks of soil and groundwater contamination. In this context, copper slag (CS), a by-product of the copper smelting process, has emerged as a fit candidate for application in concrete and mortar production. Significant improvements in the agriculture sector are resulting in more rice production, eventually resulting in abundant rice husk from developing countries, which can be turned into rice husk ash (RHA) that can act as a good pozzolanic substitute for cement. To address these challenges, the utilisation of agro-industrial by-products as alternative construction materials has gained attention as a

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sustainable and environmentally reasonable strategy.

Previous studies have demonstrated that the incorporation of CS as a partial or full replacement for fine and coarse aggregates can significantly enhance the mechanical performance of ordinary, high-strength, and high-performance concretes [1–5]. Furthermore, owing to its pozzolanic reactivity, finely ground CS has been successfully utilised as a partial cement replacement. Gopalakrishnan et al. [6] and Romy et al. [7] identified optimal replacement levels of 30 % and 20 %, respectively, for the production of high-performance mortars. The applicability of CS has also been validated in alkali-activated geopolymer concretes and in systems incorporating recycled aggregates, highlighting its versatility across diverse binder matrices [8–10]. Additionally, exposure to moderately elevated temperatures (approximately 200 °C) has been reported to further improve the strength of CS-based concretes [11]. The incorporation of CS not only enhances workability and reduces water demand – attributable to its smooth, glassy texture and relatively high specific gravity – but also contributes to increased matrix density. However, a potential concern relates to the leaching of heavy metals from the matrix. Panda et al. [12], through atomic absorption spectroscopy (AAS), reported minimal leaching, suggesting that when CS is incorporated into cementitious systems containing supplementary cementitious materials, the mobility of heavy metals can be effectively mitigated. Zain et al. [13], reported lesser leaching as per the Malaysian Environment Quality Act. RHA, due to its pozzolanic and filler nature, is a fit candidate for cement replacements in concrete and mortar matrices. Studies on replacing cement in concrete have shown that with less consumption of cement, the required performance of concrete in terms of mechanical strength and durability can be met. Past studies have shown that RHA can be substituted for cement up to 20 %, whereas increased replacement impacts the overall properties of the cementitious matrices [14–17]. The RHA is a highly effective supplementary cementitious material (SCM) due to its high amorphous silica content and strong pozzolanic reactivity. When rice husks are burned under controlled conditions between 500 and 700 °C, the resulting ash contains 80–95 % amorphous silica, which reacts with calcium hydroxide released during cement hydration to form additional calcium silicate hydrate (C–S–H), thereby refining the microstructure and improving long-term performance [18].

Studies consistently show that replacing cement with 5–20 % RHA enhances concrete strength and durability. At replacement levels around 10–15 %, compressive strength often increases due to filler effects and accelerated pozzolanic reactions [19]. RHA's high surface area and porous texture may increase water demand, but this can be compensated for by using superplasticizers or optimised fineness [20]. RHA-blended concretes exhibit significant reductions in permeability, chloride ion penetration, and sulfate attack, mainly because the formation of secondary C–S–H reduces capillary pore connectivity [21]. Additionally, RHA reduces alkali–silica reactivity by lowering calcium hydroxide content and binding alkalis within the C–S–H matrix [22]. These improvements often place RHA performance close to that of silica fume, despite being more economical and sustainable. Microstructural studies using SEM, XRD, and TGA reveal that RHA-modified concrete forms a denser and more homogeneous matrix, with refined gel pores and fewer microcracks [23]. Because rice husk is an agricultural by-product, the incorporation of RHA into concrete also contributes to reduced carbon emissions and promotes a circular economy model, especially in rice-producing regions [24]. Overall, RHA is a promising supplementary cementitious material capable of enhancing mechanical strength, durability, and sustainability when processed and used properly. Its performance depends largely on burning conditions, particle fineness, and optimized replacement levels.

Although extensive research has explored the use of CS and RHA in concrete, relatively few studies have examined their application in cement mortars, particularly in combination with cementitious replacements. The present study addresses this research gap by investigating the combined effects of dual replacements—using CS as a fine

aggregate substitute and RHA as partial cement replacement. Building upon the findings of Al-Jabri et al. [25] and Sensale [14], who identified an optimal CS replacement range of 30–50 % in concrete and cement mortars and 10–20 % RHA from other studies, the current study investigates whether comparable performance can be achieved in cement mortars incorporating supplementary cementitious material such as RHA. Furthermore, the study aims to develop empirical relationships for predicting compressive and flexural strengths, which remain scarcely reported in existing literature. In this investigation, cement was replaced with RHA at levels of 10 % and 20 %, while CS replaced fine aggregates at 10 %, 30 %, and 50 %. The mechanical behaviour of the mortars was evaluated through compressive and flexural strength tests, and ultrasonic pulse velocity (UPV) testing was performed only on flexural specimens to assess the matrix quality and integrity of the developed mixes.

2. Experimental programme

2.1. Materials

The preparation of cement mortar followed the procedures outlined in IS 2250:1981 [26], while the fine aggregate used conformed to the specifications of IS 2116:1980 [27]. A cement-to-sand ratio of 1:3 and a water-cement ratio of 0.5 were adopted. To establish the optimum mix design, several mortar mixes were prepared incorporating partial replacement of cement with RHA and replacement of fine aggregate with CS using a volume batching technique. Superplasticiser was not used. The physical and chemical properties of CS and RHA are reported in the author's earlier work [28] and are therefore not reproduced here for brevity. Mortar mix notation is provided in Table 1.

Table 1 presents the mix proportions adopted in this study. Each mortar mix is presented as N-R-P-C, where N stands for the percentage of RHA replacement, R for RHA, P for the percentage replacement of CS and C for CS. This resulted in a series of mixes- 10R10C, 10R30C, 10R50C, 20R10C, 20R30C, 20R50C- designed to evaluate the influence of varying replacement levels on the properties of cement mortar. The control mix (CM) was prepared using 100 % cement and 100 % manufactured sand.

For compressive strength testing, 50 × 50 mm cubes were prepared in accordance with IS 4031 Part 7 [29]. Three cubes were cast for each mix and tested at curing ages of 7 and 28 days, following the procedure specified in BS EN 1015–11 [36]. The weight of each specimen was recorded before testing. Flexural strength was evaluated using 40 × 40 × 160 mm prism specimens prepared in accordance with BS EN 1015–11 [36]. UPV measurements were performed on the flexural prisms following the guidelines of IS 516 (Part 1) [30].

2.2. Experimental test methods

2.2.1. Mortar flow test

The flow test for mortar was conducted in accordance with the procedures specified in IS 2250 [26] and IS 4031 [29], as illustrated in Fig. 1. For masonry mortars, these standards recommend a flow range of 110–115 mm to ensure adequate workability. The results obtained for all mixes in this study fell within the specified range, confirming the

Table 1
Mix design and designation.

Mix Designation	Cement (%)	RHA (%)	Sand (%)	CS (%)
Control (CM)	100	0	100	0
10R10C	90	10	90	10
10R30C	90	10	70	30
10R50C	90	10	50	50
20R10C	80	20	90	10
20R30C	80	20	70	30
20R50C	80	20	50	50

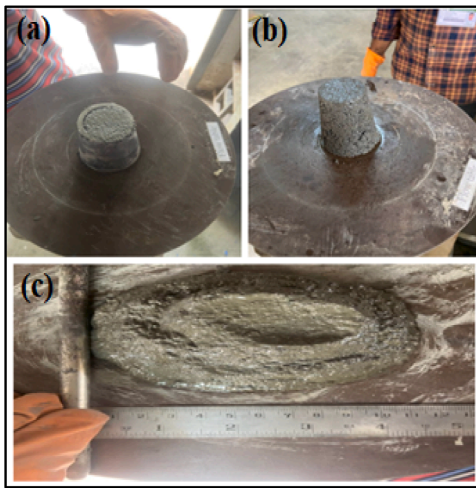


Fig. 1. Mortar flow test on specimens.

satisfactory workability and practical applicability of the prepared mortars.

2.2.2. UPV test on flexural prism

Before conducting the flexural strength test, the mortar prism specimens were subjected to the UPV test to evaluate the quality, uniformity, and homogeneity of the mixes incorporating partial replacement of cement with RHA and fine aggregate with CS. A direct method was adopted wherein the two probes were held against the finished sides of the specimens so that the waves travel straight and are received at the end. A 53 Hz frequency was used. The experimental setup used for the UPV test is illustrated in Fig. 2 as per IS 516 (Part 1) [30].

2.2.3. Compressive strength

The compressive strength test on mortar cubes was carried out in accordance with BS EN 1015–11 [43] using a calibrated compression testing machine capable of applying load at a controlled rate of 50 N/s, as shown in Fig. 3(a). The load was applied smoothly and continuously until failure, and the maximum load at failure was recorded. The compressive strength was determined by dividing the maximum load by the cross-sectional area of the specimen. Individual results were recorded to the nearest 0.05 N/mm², and the average strength of three specimens was reported to the nearest 0.1 N/mm² for both 7-day and 28-day curing periods.

2.2.4. Flexural strength

The flexural strength test on mortar prisms was determined at 7- and 28-days following BS EN 1015–11 [42]. The setup comprised a simple



Fig. 2. UPV (Direct) test on flexural specimen.

test rig prepared using angles welded at the ends, as shown in Fig. 3b, maintaining centre-to-centre spacing of 100 ± 0.5 mm and a small rod of approximately 10 mm in diameter was used to apply load from the machine. Tests were performed using a universal testing machine capable of applying a uniform load at a rate of 10–50 N/s, ensuring failure occurred within 30–90 s, and the fractured specimen is illustrated in Fig. 3(b).

3. Results and discussion

3.1. Mortar flow

The mortar flow test results indicate that all mixes achieved flow values within the acceptable range recommended by IS standards (110–115 mm), as shown in Fig. 4(a), confirming adequate workability for masonry applications. The CM recorded a flow of 110 mm, while the mix containing 10 % RHA and 10 % CS (10R10C) exhibited the highest workability at 115 mm. Increasing CS content to 30 % and 50 % slightly reduced the flow to 108 mm and 106 mm, respectively, likely due to the angularity and rough texture of the slag particles. Other constituents of the mortar, such as superplasticiser, RHA, and w/c ratio, can affect the flow. In the case of Arpitha et al. [31] studies, flow is reduced in a mix containing >50 % CS, and this is in agreement with the observations in the current study. Similarly, Edwin et al. [7] reported an increase in workability as the CS percentage increases. Mixes with 20 % RHA showed progressively lower flow values, decreasing from 110 mm at 10 % CS to 103 mm at 50 % CS, reflecting the higher water demand and fineness of RHA. Overall, limited substitution with 10 % RHA and 10 % CS enhanced workability compared to 20 % RHA, whereas higher replacement levels reduced flow, though all the values remained within permissible limits.

The bulk density results revealed a clear influence of RHA and CS on the density of mortar specimens, as shown in Fig. 4(b). These results confirm that CS, owing to its higher specific gravity, increased the specimen weight and density, whereas higher RHA content led to a reduction due to its lower density and greater porosity. These variations also align with the UPV trends, where denser and heavier mixes exhibited higher pulse velocities, reflecting improved microstructural quality, which will be discussed in the subsequent sections.

3.2. Compressive and flexural strength

The compressive strength results demonstrate a clear enhancement with the incorporation of RHA and CS compared to the CM, as shown in Fig. 5(a). At 7 days, the CM achieved 9.07 MPa, while mixes containing 10 % RHA and increasing levels of CS exhibited higher early strengths, reaching up to 18.26 MPa. The maximum 28-day strength was recorded for the mix with 10 % RHA and 50 % CS, attaining 38 MPa, a substantial improvement over the CM (18.53 MPa). Mixes with 20 % RHA displayed lower early strengths, particularly at 7 days, but showed notable gains at 28 days, reaching 25.5 MPa and 28.3 MPa for 30 % and 50 % CS replacement, respectively. Overall, the results indicate that combining 10 % RHA with higher proportions of CS yields the most significant improvement in compressive strength, whereas higher RHA content (20 %) reduces early strength but ensures satisfactory long-term performance. In line with the findings of the current study, substitution of CS for fine aggregates results in a substantial increment in compressive strength [6,25,32]; however, Resende et al. [33] report an opposite trend of strength decrement due to CS, and this could be due to the inadequate hydration within the mortar matrix, as hardening was delayed. On the other hand, incorporation of 20 % RHA and 10 % CS is the best mix compared to other mixes with 20 % RHA. This is primarily due to the balance between pozzolanic reaction rate, particle packing efficiency, and workability requirements and at 10 % RHA, according to Mehta and Monteiro [18], Zhang and Malhotra [19], amount of RHA available is well matched to the quantity of calcium hydroxide released

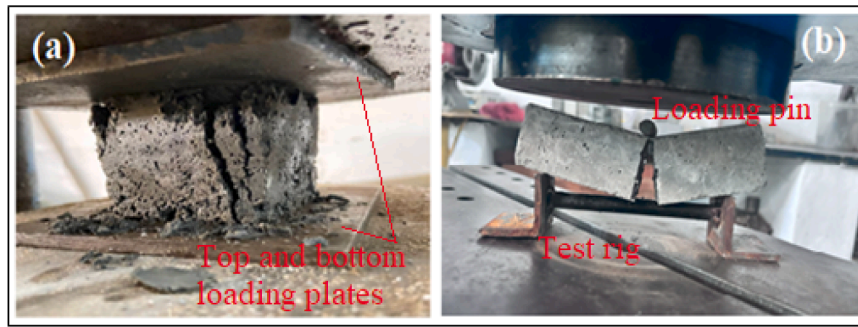


Fig. 3. Compressive strength test on mortar.

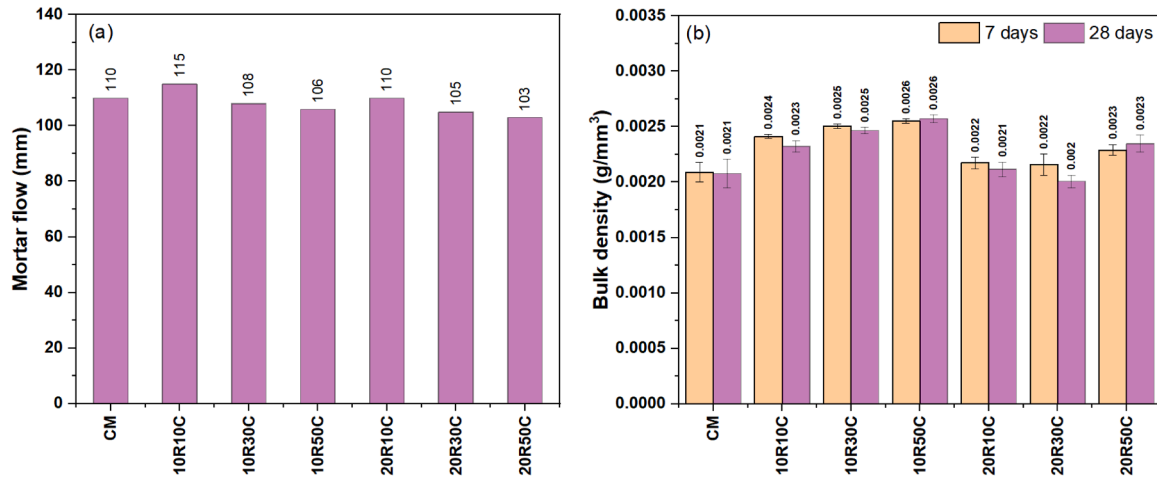


Fig. 4. (a) Flow test on cement mortar specimens; and (b) Measured cube weights of all mixes.

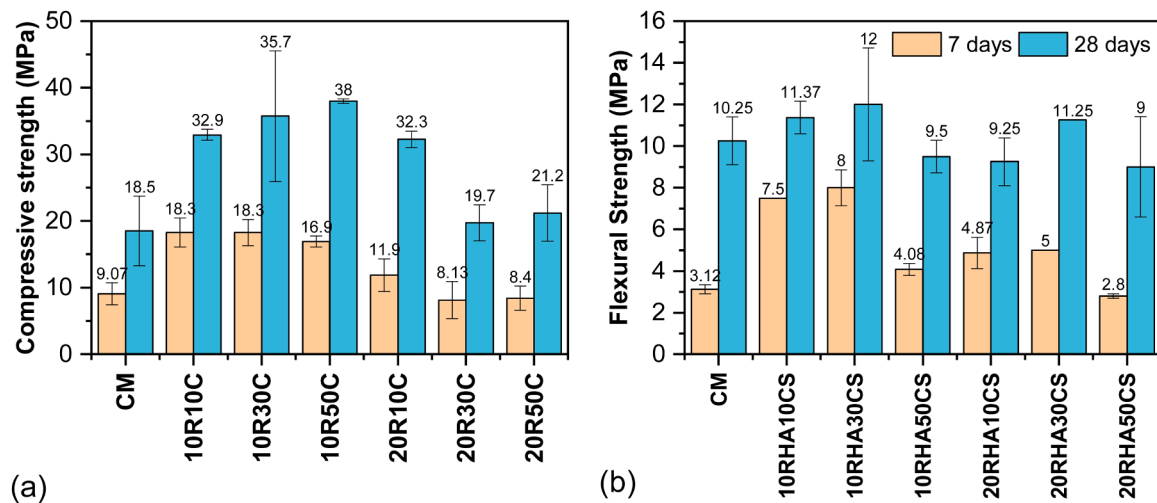


Fig. 5. Evolution of mortar strengths: (a) compressive strength; and (b) flexural strength.

during normal cement hydration, resulting in efficient pozzolanic conversion without leaving excess unreacted ash or calcium hydroxide. On another level, the microfiller effect of finely ground RHA could improve the surrounding matrix, making it dense; however, when RHA increases >10 %, a high water demand will possibly contribute to lower mechanical strengths, as seen in Fig. 5(a) and (b). Hence, 10 % RHA in combination with 50 % CS has shown better compressive strength, indicating an optimum synergy exists between pozzolanic activity and water demand.

The flexural strength results reveal a marked improvement in performance with the incorporation of RHA and CS compared to the CM, as shown in Fig. 5(b). At 7 days, the CM achieved only 3.12 MPa, whereas mixes containing 10 % RHA and varying CS levels exhibited higher values, with 10R30C reaching 8 MPa. The highest 28-day flexural strength was also recorded for 10R30C at 12 MPa, nearly 1.17 times greater than the CM (10.25 MPa). Similarly, 10R10C and 10R50C exhibited strong performance, achieving values above 11 MPa at 28 days. Mixes with 20 % RHA showed lower early strengths, ranging from

2.8 to 5 MPa at 7 days, but attained satisfactory 28-day strengths between 9 and 11.25 MPa. Overall, these results suggest that partial replacement of cement with 10 % RHA, combined with 30–50 % CS, yields the most favourable effect on flexural strength, while higher RHA content (20 %) reduces early performance but maintains adequate long-term strength.

Concrete containing 10 % RHA typically exhibits equal or higher flexural strength compared to conventional concrete. This improvement is linked to several microstructural mechanisms that get optimised at this replacement level. First, the pozzolanic reaction between the amorphous silica in RHA and the calcium hydroxide released during cement hydration forms additional calcium silicate hydrate (C–S–H). This secondary C–S–H enhances the bond between the cement matrix and fine aggregates, which is crucial for resisting tensile stresses developed during flexure [18,19].

Second, finely ground RHA particles act as microfillers, improving the packing density and reducing microvoids along potential crack paths. This results in a denser interfacial transition zone (ITZ), which increases crack resistance and enhances flexural performance [20]. A refined ITZ is particularly beneficial for flexural strength because flexural failure typically initiates at the ITZ or at weak points in the tensile zone. Experimental studies consistently report that concrete incorporating around 10 % RHA shows a 5–20 % increase in flexural strength, depending on curing conditions, fineness of RHA, and mix design. Ganesan et al. [34] observed that RHA concretes at 10 % replacement exhibited a higher modulus of rupture due to improved microstructure and reduced pore connectivity. Similarly, Rukzon & Chindaprasit [21] reported enhanced flexural performance in high-strength mixes containing RHA due to the reduced calcium hydroxide content and denser C–S–H network. However, when RHA content exceeds 15–20 %, the flexural strength may start to decline due to increased water demand, reduced cementitious material, and potential compaction issues [34].

3.3. Ultrasonic pulse velocity

The UPV test results show that all mortar mixes fall within the range of 4900–5500 m/s, indicating good quality and dense microstructure across all the specimens. The CM achieved 5232 m/s at 7 days and 5442 m/s at 28 days, while the remaining mix values are presented in Fig. 6. With 10 % RHA and varying CS replacements, a consistent improvement was observed: 10R30C and 10R50C reached 5373.67 m/s and 5150.33 m/s at 28 days, respectively, reflecting enhanced densification due to the pozzolanic activity of RHA and the filler effect of CS. The highest UPV value was recorded for 20R30C at 28 days (5471.67 m/s), exceeding that of the CM, which suggests superior internal quality and

reduced voids. However, the 20R50C mix exhibited slightly lower values (4961 m/s at 7 days and 5424.67 m/s at 28 days), indicating that excessive replacement may reduce uniformity at early ages but still results in satisfactory density over time. Overall, the findings confirm that incorporating RHA and CS enhances the compactness and homogeneity of mortar, with optimum performance achieved at 10 % RHA with 30–50 % CS and 20 % RHA with 30 % CS. Tiwary et al. [35] show similar results with the incorporation of fly ash at 20 % and CS at 30 %. A relationship between flexural strength at 7 and 28 days is reported in Fig. 7(a) and 7(b), showing a moderate correlation. Fig. 7(a) shows an exponentially increasing trend, indicating that at early strength development, the matrix is still in its early stage, with hydration still in progress. Whereas, RHA at twenty-eight days is sufficient but not completely hydrated, resulting in better performance.

The moderate R^2 obtained in this study reflects the inherent variability of flexural strength in cementitious composites. Strength development could be influenced by multiple microstructural and experimental factors, such as pore structure, interfacial transition zone quality, and specimen surface defects, which were not explicitly included in the regression model. Flexural strength, in particular, can exhibit higher experimental scatter than compressive strength. Despite these limitations, the model captures the trend.

3.4. Prediction of compressive and flexural strengths

A multiple regression analysis was conducted to establish the relationship between flexural strength and key influencing parameters. The independent variables considered included curing age, RHA content, and CS percentage. The resulting regression equation for the overall flexural strength model is presented in Eq. (1). The model achieved an overall R^2 value of 0.73, indicating good predictive accuracy. The terms in the equations have been defined elsewhere, except T, which stands for the 7th and 28th days strength. A summary of the developed model is provided in Table 2. As observed, RHA content and curing time were highly significant, whereas CS content showed a p-value of 0.58, indicating a relatively lower influence. Similarly, a multiple linear regression analysis was performed to develop another flexural strength model, which yielded an overall R^2 value of 0.76, as shown in Eq. (2), with its summary presented in Table 3. The model parameters exhibited p-values similar to those obtained for the compressive strength model. A comparison of data from the current study and from the data available in the literature is presented in Figs. 8 and 9, and the residuals from the statistical analysis in Fig. 10.

$$f_c = 22.42 + (0.77 \times T) - (0.85 \times RHA) - (0.032 \times CS) \quad (1)$$

$$f_t = 5.1 + (0.28 \times T) - (0.21 \times RHA) - (0.024 \times CS) + (0.00036 \times UPV) \quad (2)$$

A set of compressive and flexural strength data was selected from the literature on cement mortars incorporating CS [6, 36–41]. The compressive strength values from these studies were used to validate the equations developed in the present work. Overall, the findings indicate that the comprehensive model provides a more robust and statistically reliable prediction of compressive strength development.

The comparison plot of predicted versus experimental compressive strength shows good correlation, with most points closely following the ideal 1:1 line (black). The overall clustering of points around the reference line demonstrates the model's accuracy in capturing the flexural strength behaviour, especially Ma et al. [11], Bhoi et al. [36] and Rezaeian et al. [38]. The few deviations beyond the ± 20 % limits confirm that the prediction method is reliable. In the case of the flexural strength prediction equation, most of the data from the current study fall within the prediction lines with the Ameri et al. [37] and most of Bhoi et al. [36] beyond the prediction lines.

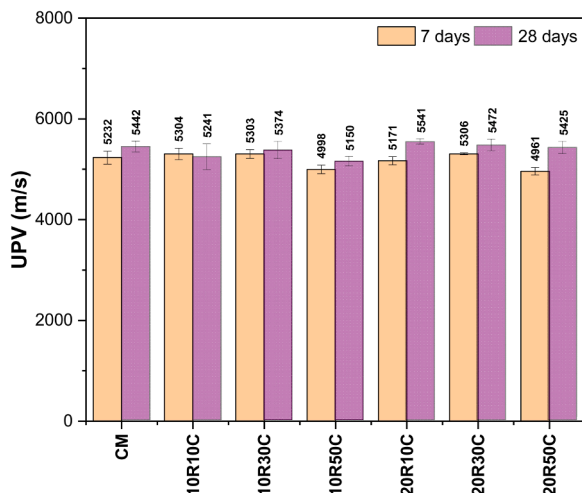


Fig. 6. UPV values of flexure specimens.

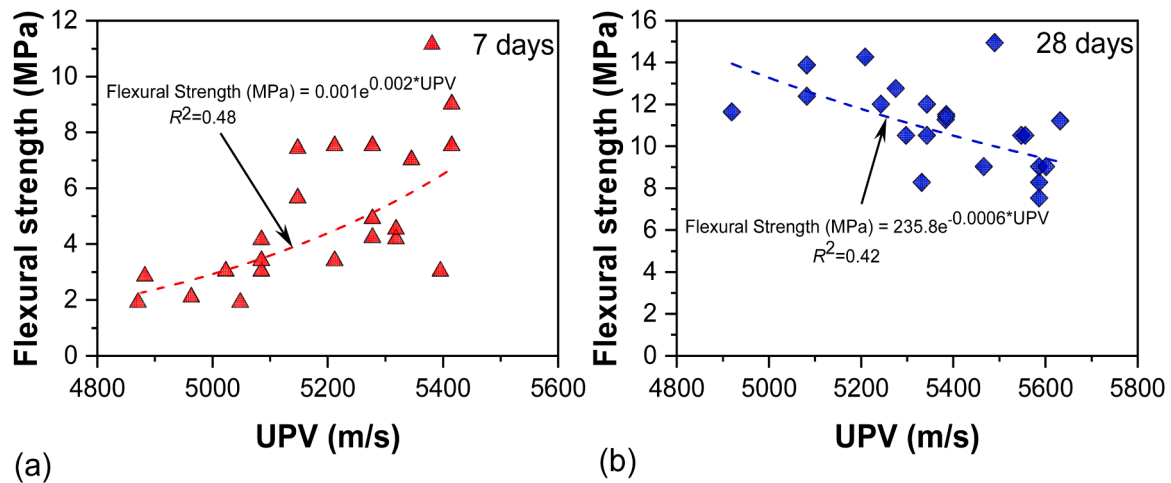


Fig. 7. Correlation between flexural strength and UPV values: (a) 7 days, and (b) 28 days.

Table 2

Summary of the multiple linear regression model for compressive strength using CS and RHA.

Independent variables	Coefficient	Std. error	t-value	p-value
Constant	22.42	3.68	6.09	8.16E-07
T	0.77	0.08	8.77	5E-10
RHA	-0.85	0.18	-4.50	8.27E-05
CS	-0.032	0.05	-0.55	0.58

Table 3

Summary of the multiple linear regression model for flexural strength using CS and RHA.

Independent variables	Coefficient	Std. error	t-value	p-value
Constant	5.10	9.32	0.54	0.58
T	0.28	0.03	8.62	9.8E-10
RHA	-0.21	0.06	-3.46	0.001
CS	-0.024	0.02	-1.19	0.24
UPV	0.00036	0.001	0.20	0.84

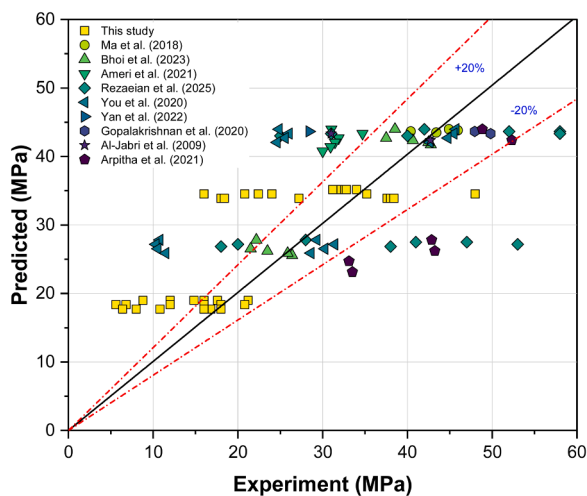


Fig. 8. Comparison of compression strength with current study and others.

3.5. Image analysis

Fig. 11(a-g) presents the cross-sectional views of $40 \times 40 \times 160$ mm flexural specimens containing varying proportions of CS and RHA. The

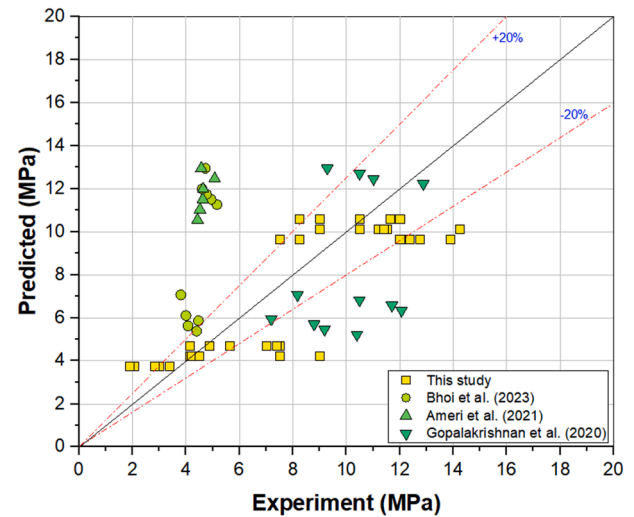


Fig. 9. Comparison of flexural strength with current study and data from literature.

cross-sectional images illustrate the effects of these supplementary materials on the compactness, texture, and pore distribution within the cementitious matrix. Fig. 10(a) shows CM, without CS and RHA, exhibiting a relatively dense and uniform surface with minimal visible pores. Fig. 10(b) depicts a specimen with 10 % fine aggregate replacement by CS, showing increased surface roughness and slightly more noticeable voids.

Fig. 10(c) and 10(d) illustrate differences in the distribution of CS aggregates, with Fig. 10(d) showing a denser arrangement compared to Fig. 10(c). The presence of RHA is not directly captured in the images; however, a slight black tinge in Fig. 10(e-g) indicates its presence. Specimens 20R10C and 20R30C, containing 10 % and 30 % CS, respectively, show noticeable changes in CS aggregate distribution, whereas 20R50C in Fig. 10(g) exhibits a less dense CS distribution, likely due to non-uniform mixing. These images suggest that copper slag contributes to increased surface coarseness and a darker hue, while RHA produces a subtle colour change in the matrix.

To further investigate the microstructure, binary images were analysed as shown in Fig. 12(a-g) using ImageJ software[42]. In these images, black regions represent the mortar matrix with RHA and without CS, while white regions indicate a combination of CS aggregates and voids. The density of CS aggregate distribution can be obtained fairly accurately with these binary images, especially in Fig. 11(d, f and g). The

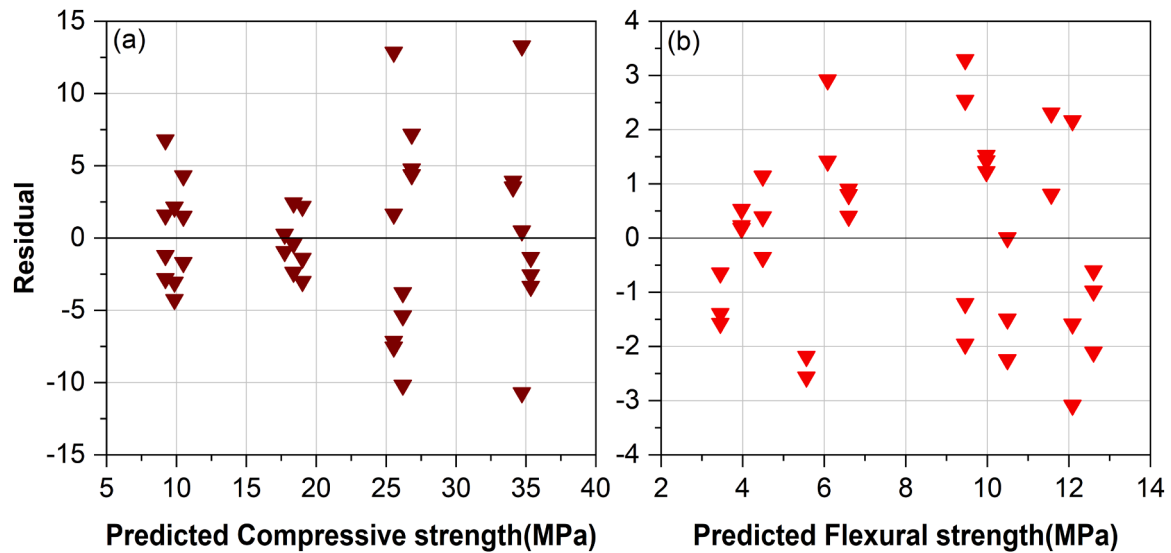


Fig. 10. Comparison of predicted compressive and flexural strengths and residuals for the developed model.

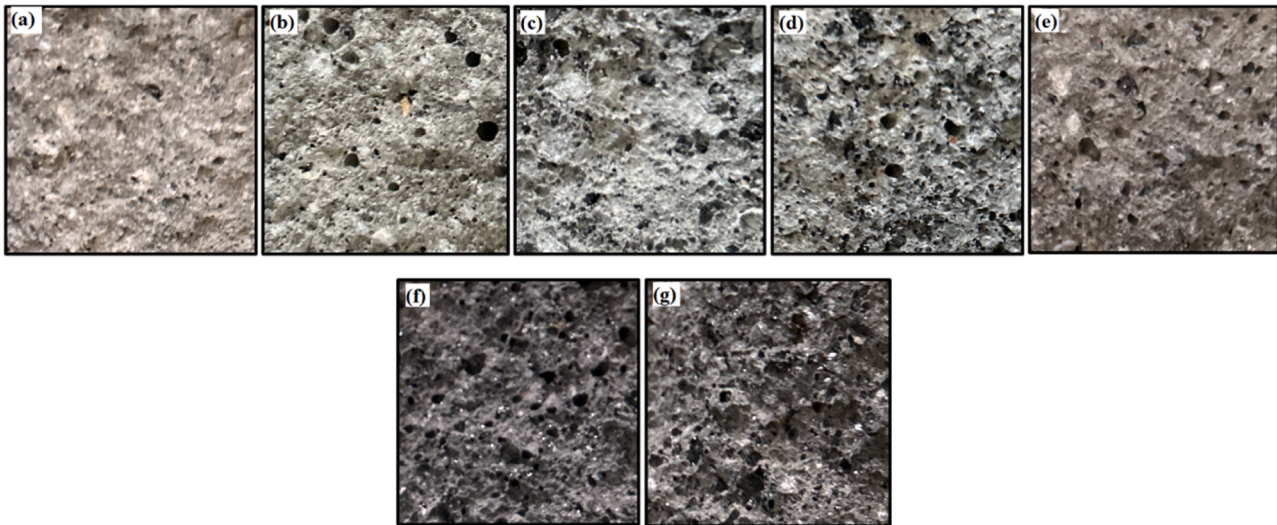


Fig. 11. Cross-section of flexure specimens.

Fig. 13(a) and 13(b) present plots of grey value versus distance, capturing voids, RHA-cement, and CS distributions. The plots mainly highlight the distribution of CS aggregates across the cross-section. The CM specimen cross-section shows the highest grey values, reflecting a uniformly black matrix with no CS aggregates. For mixes containing CS, grey values were plotted according to the percentage of CS content. In Fig. 13(a), most of the curve for the M5 mix lies below the others, indicating higher CS content, while M8 and M10 show almost identical grey values, despite expectations of a similar trend, likely due to the random distribution of CS aggregates.

4. Limitations and future recommendations

The experimental campaign was limited to two curing ages (7 and 28 days), which restricts the ability to fully capture the long-term strength development, especially with RHA in the current study. The regression models were developed using a linear formulation, which simplifies the inherently nonlinear behaviour of the tested ranges. Flexural strength results exhibited moderate scatter, which is inherent to tensile-based tests and may have influenced the coefficient of determination (R^2),

particularly for UPV–flexural strength correlations. Image analysis was limited to two-dimensional cross-sectional observations, providing qualitative insight into CS distribution but not fully capturing voids or microstructure.

Future studies should include additional curing ages to better understand long-term strength gain and the delayed pozzolanic contribution of RHA. Machine learning approaches (e.g., nonlinear regression, ANN, or hybrid models) may be employed to improve strength prediction accuracy and capture complex interactions between material parameters, provided more data is obtained. To characterise the microstructure, techniques such as SEM and XRD can be employed. Life cycle assessment (LCA) and carbon footprint analysis can be incorporated to quantitatively assess the environmental benefits of using RHA and CS in cement mortar systems.

5. Conclusions

This study demonstrated that the combined use of rice husk ash (RHA) and copper slag (CS) significantly affects both the fresh and hardened properties of cement mortar. Flow tests confirmed that all

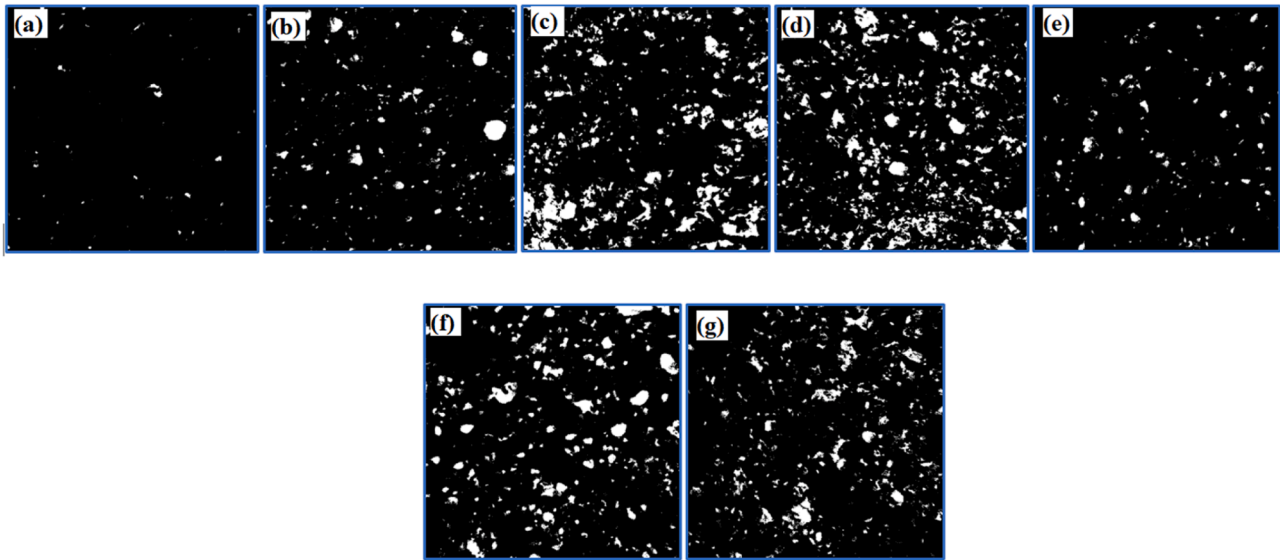


Fig. 12. Cross-section of flexure specimens-binary images.

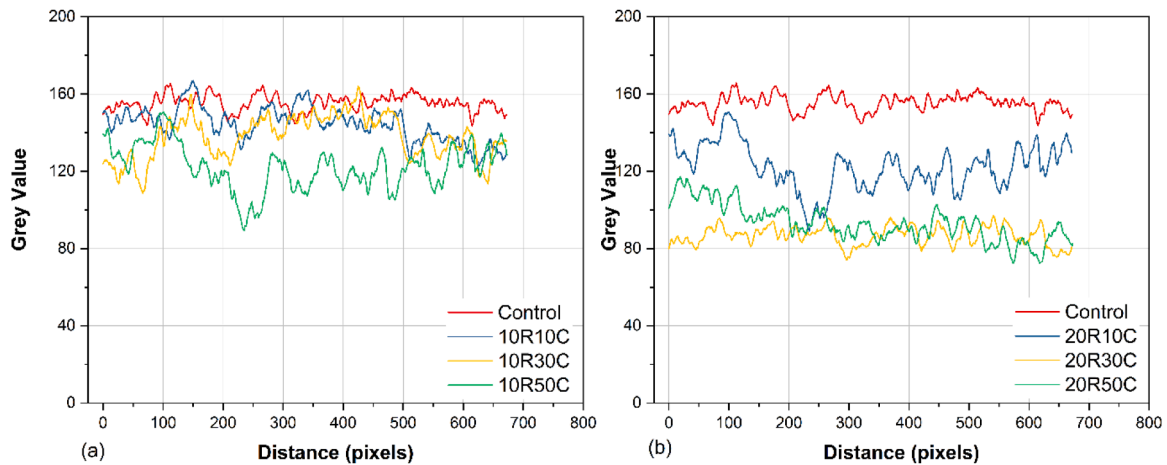


Fig. 13. Comparison of grey value and pixels across the cross-section of all specimens.

mixes show reduced and increased water demand. Bulk weight trends showed that CS increased the density of cubes and prisms due to its higher specific gravity, whereas higher RHA replacement reduced weight because of its lower density and higher porosity. Some of the important findings have been summarised in the following section.

1. All mortar mixes satisfied workability requirements (110–115 mm flow), with the 10R10C mix showing the highest flow, representing an approximate 5 % increase compared to the control mix.
2. Replacement of manufactured sand with copper slag increased specimen density by approximately 4–8 %, owing to its higher specific gravity, while increasing RHA content reduced specimen weight by approximately 3–6 % due to its lower density and porous structure.
3. The maximum 28-day compressive strength of 38 MPa, achieved with 10 % RHA and 50 % CS, represents a 105 % increase compared to the control mix (18.53 MPa).
4. Flexural strength improved significantly with RHA–CS incorporation, with 10 % RHA and 30 % CS attaining 12 MPa at 28 days, corresponding to a 17 % increase over the control mix (10.25 MPa).
5. Ultrasonic pulse velocity values for all mixes ranged from 4900 to 5500 m/s, indicating good to excellent quality mortar; the highest UPV recorded for 20 % RHA and 30 % CS showed an approximate 4–5 % increase over the control mix at 28 days. A moderate correlation between UPV and compressive, flexural strengths revealed an R^2 of 0.42 and 0.48, respectively.
6. Multiple linear regression analysis confirmed curing age as the dominant parameter governing strength development, with adjusted R^2 values of 0.73 for compressive strength and 0.76 for flexural strength, indicating that 73–76 % of strength variability was successfully captured by the models.
7. The close agreement between predicted and experimental values, with most data points lying within ± 20 % prediction limits, confirms the robustness of the developed regression equations despite the inherent variability of cementitious composites.
8. Image-based analysis verified a uniform and random distribution of copper slag particles, supporting improved particle packing and efficient stress transfer within the mortar matrix.
9. Overall, the incorporation of 10 % RHA combined with 30–50 % CS offers the most balanced performance, delivering 15–105 % improvements in mechanical properties while maintaining workability.

The study demonstrates that the synergistic use of RHA and CS offers a technically viable and sustainable approach for enhancing cement

mortar performance, supporting the utilisation of industrial by-products in environmentally responsible construction practices.

Data availability

All data that support the findings of this study are included within the article.

CRediT authorship contribution statement

C.D. Vallabha: Writing – original draft, Software, Resources, Methodology, Investigation, Formal analysis, Conceptualization. **Naveen Revanna:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Arun B.R.:** Writing – review & editing, Supervision, Project administration, Investigation, Formal analysis, Conceptualization. **Hemanth Kumar Veeraiah:** Writing – review & editing, Visualization, Supervision, Software, Investigation, Formal analysis. **Shiva Prashanth Kumar Kodicherla:** Writing – review & editing, Visualization, Validation, Software, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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